The empirical recurrence for OEIS A255025, recovered from its tail

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Abstract

OEIS A255025 records "Empirical recurrence of order 71" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 106$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 74$. Its last coefficient is $c_{71} = 4 \neq 0$, so no further tail satisfies anything shorter; any order-71 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A255025 is "Number of (n+2)X(6+2) 0..1 arrays with no 3x3 subblock diagonal sum 0 or 1 and no antidiagonal sum 0 or 1 and no row sum 0 or 1 and no column sum 0 or 1.". It has offset 1 and begins

$$1306, 8338, 60237, 391926, 2547757, 16789556, 110131505, 722619025, \ \ldots$$

The entry states:

> Empirical recurrence of order 71 (see link above)

As of the "Last modified" line on the live entry (Jun 02 2025 11:36:17, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 106$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 106$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 71: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 71.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 212$ exact terms from $n = 3$ on, it returns order 71 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{71} c_i \, a(n-i),$$

c_1	1	c_{19}	−12344809	c_{37}	−240921735	c_{55}	−5343415
c_2	16	c_{20}	293828	c_{38}	−60874573	c_{56}	1932647
c_3	82	c_{21}	29574010	c_{39}	82422696	c_{57}	−140432
c_4	372	c_{22}	18444680	c_{40}	−84170974	c_{58}	−9730
c_5	420	c_{23}	−10085249	c_{41}	311567049	c_{59}	158849
c_6	−1824	c_{24}	−9847451	c_{42}	−166351908	c_{60}	−144988
c_7	−10711	c_{25}	−46503477	c_{43}	177380254	c_{61}	69529
c_8	−25527	c_{26}	−63383581	c_{44}	−60332045	c_{62}	−37326
c_9	25103	c_{27}	40314409	c_{45}	−72362554	c_{63}	14156
c_{10}	218652	c_{28}	32483704	c_{46}	101161085	c_{64}	−3426
c_{11}	211377	c_{29}	55295268	c_{47}	−156688641	c_{65}	939
c_{12}	−576454	c_{30}	152288762	c_{48}	155712965	c_{66}	1367
c_{13}	−1605679	c_{31}	−41410989	c_{49}	−129434276	c_{67}	−866
c_{14}	−266995	c_{32}	−54142094	c_{50}	108031833	c_{68}	132
c_{15}	4337025	c_{33}	−171478377	c_{51}	−70710813	c_{69}	4
c_{16}	4812726	c_{34}	−305657998	c_{52}	42806311	c_{70}	−20
c_{17}	−2110119	c_{35}	−308595535	c_{53}	−25012842	c_{71}	4
c_{18}	−8042065	c_{36}	−225772958	c_{54}	11081702		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 74$, and it is the recurrence OEIS A255025 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{71} - c_1 x^{71-1} - \cdots - c_{71}$. Its constant term is $-c_{71} = -4$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$

vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 71 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 71, so $\deg q = 71$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 71, so $q = q_{\min}$. □

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 18 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 71, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 4.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-71 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A255025.

[2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)

[5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.